

Algorithms for Data Science

Lecture 8: Clustering, Model Evaluation, and Model Optimization



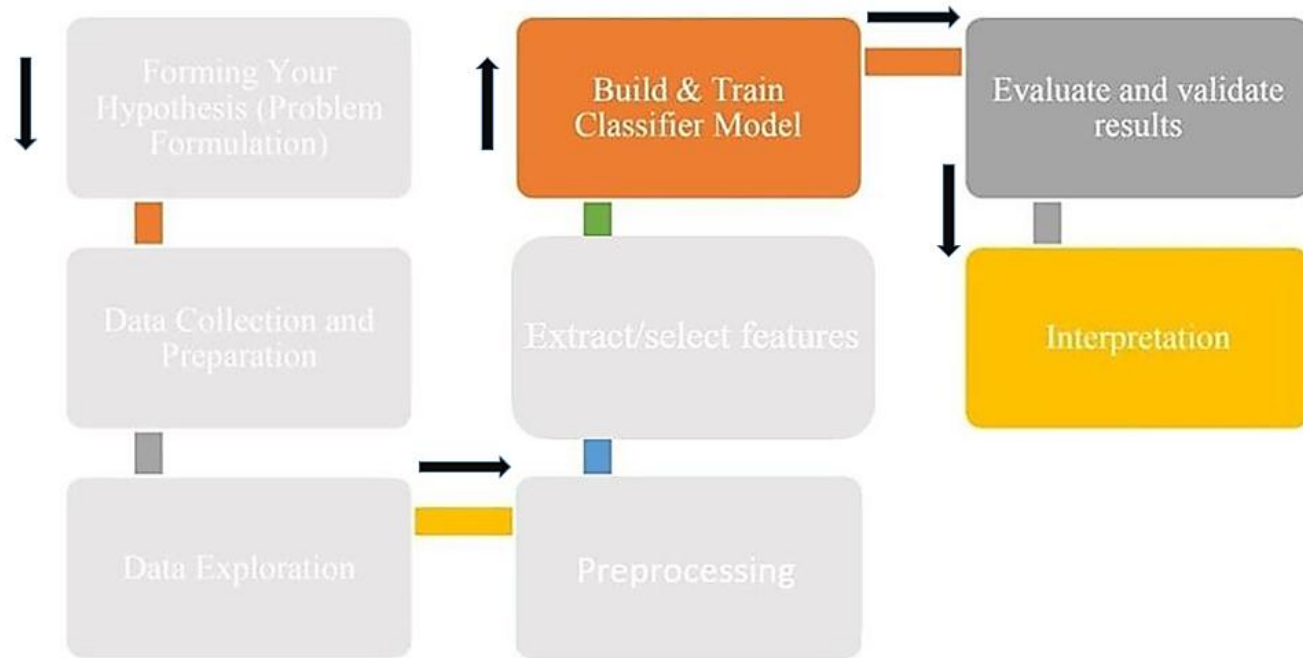


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The Data Science Pipeline



Clustering

- Clustering is an unsupervised learning technique that discovers natural groupings within data without using labeled examples.
- **Real-World Applications:** Customer segmentation, document organization, image compression, anomaly detection.
- Clustering is not a single algorithm but a family of techniques, each with distinct mathematical foundations and suitable applications. The choice of clustering method depends on the data characteristics, cluster shapes, scalability requirements, and domain constraints.

Major Clustering Categories

Category	Core Principle	Representative Algorithms
Partitional	Divide data into non-overlapping subsets	K-Means
Hierarchical	Build nested tree of clusters	Agglomerative

K-Means

- K-Means is a centroid-based partitional clustering algorithm that partitions n observations into K disjoint clusters, where each observation belongs to the cluster with the nearest mean (centroid).
- **K-Means include main four steps:**
 - **Step 1 – Initialization:** Randomly select K data points from the dataset as initial centroids.
 - **Step 2:** Compute the Euclidean distance from each data point to every centroid. Assign each point to its nearest centroid. This creates a hard partition of the data.
 - **Step 3:** For each cluster, recalculate the centroid as the arithmetic mean of all points assigned to that cluster.
 - **Step 4 – Convergence Check:** Compare new centroids to previous centroids. If the maximum change in any centroid falls below a tolerance threshold ϵ (typically $1e-4$), terminate. Otherwise, return to Step 2.



K-Means Clustering- Example

Problem: Cluster six points into $K=2$ clusters using K-Means.

Point	X	Y
A	2	3
B	3	3
C	8	8
D	9	9
E	10	7
F	11	6

Initial Centroids (Randomly Selected):

- Centroid 1 (Red) = Point A = (2, 3)
- Centroid 2 (Blue) = Point D = (9, 9)



K-Means Clustering- Example

Iteration 1: Assignment Step (Distance Calculation Table)

Point	Distance to Red (2,3)	Distance to Blue (9,9)	Closer To	Assignment
A (2,3)	$\sqrt{[(2-2)^2 + (3-3)^2]} = \sqrt{0} = \mathbf{0}$	$\sqrt{[(2-9)^2 + (3-9)^2]} = \sqrt{(49+36)} = \sqrt{85} = \mathbf{9.22}$	Red	Cluster 1
B (3,3)	$\sqrt{[(3-2)^2 + (3-3)^2]} = \sqrt{1} = \mathbf{1}$	$\sqrt{[(3-9)^2 + (3-9)^2]} = \sqrt{(36+36)} = \sqrt{72} = \mathbf{8.49}$	Red	Cluster 1
C (8,8)	$\sqrt{[(8-2)^2 + (8-3)^2]} = \sqrt{(36+25)} = \sqrt{61} = \mathbf{7.81}$	$\sqrt{[(8-9)^2 + (8-9)^2]} = \sqrt{(1+1)} = \sqrt{2} = \mathbf{1.41}$	Blue	Cluster 2
D (9,9)	$\sqrt{[(9-2)^2 + (9-3)^2]} = \sqrt{(49+36)} = \sqrt{85} = \mathbf{9.22}$	$\sqrt{[(9-9)^2 + (9-9)^2]} = \sqrt{0} = \mathbf{0}$	Blue	Cluster 2
E (10,7)	$\sqrt{[(10-2)^2 + (7-3)^2]} = \sqrt{(64+16)} = \sqrt{80} = \mathbf{8.94}$	$\sqrt{[(10-9)^2 + (7-9)^2]} = \sqrt{(1+4)} = \sqrt{5} = \mathbf{2.24}$	Blue	Cluster 2
F (11,6)	$\sqrt{[(11-2)^2 + (6-3)^2]} = \sqrt{(81+9)} = \sqrt{90} = \mathbf{9.49}$	$\sqrt{[(11-9)^2 + (6-9)^2]} = \sqrt{(4+9)} = \sqrt{13} = \mathbf{3.61}$	Blue	Cluster 2

K-Means Clustering- Example

Assignment Summary Table

Cluster	Points Assigned	Count
Cluster 1 (Red)	A, B	2
Cluster 2 (Blue)	C, D, E, F	4



K-Means Clustering- Example

Iteration 1: Update Step (New Centroids)

Centroid 1 Calculation Table (Points A and B)

Point	X	Y
A	2	3
B	3	3
Sum	$2 + 3 = 5$	$3 + 3 = 6$
Centroid 1	$5 \div 2 = 2.5$	$6 \div 2 = 3$

New Red Centroid = (2.5, 3)

Centroid 2 Calculation Table (Points C, D, E, F)

Point	X	Y
C	8	8
D	9	9
E	10	7
F	11	6
Sum	$8+9+10+11 = 38$	$8+9+7+6 = 30$
Centroid 2	$38 \div 4 = 9.5$	$30 \div 4 = 7.5$

New Blue Centroid = (9.5, 7.5)

K-Means Clustering- Example

Iteration 2: Assignment Step (Using Updated Centroids)- Red = (2.5, 3) and Blue = (9.5, 7.5)

Point	Distance to Red (2.5,3)	Distance to Blue (9.5,7.5)	Closer To	Assignment
A (2,3)	$\sqrt{[(2-2.5)^2 + (3-3)^2]} = \sqrt{0.25} = \mathbf{0.50}$	$\sqrt{[(2-9.5)^2 + (3-7.5)^2]} = \sqrt{(56.25+20.25)} = \sqrt{76.5} = \mathbf{8.75}$	Red	Cluster 1
B (3,3)	$\sqrt{[(3-2.5)^2 + (3-3)^2]} = \sqrt{0.25} = \mathbf{0.50}$	$\sqrt{[(3-9.5)^2 + (3-7.5)^2]} = \sqrt{(42.25+20.25)} = \sqrt{62.5} = \mathbf{7.91}$	Red	Cluster 1
C (8,8)	$\sqrt{[(8-2.5)^2 + (8-3)^2]} = \sqrt{(30.25+25)} = \sqrt{55.25} = \mathbf{7.43}$	$\sqrt{[(8-9.5)^2 + (8-7.5)^2]} = \sqrt{(2.25+0.25)} = \sqrt{2.5} = \mathbf{1.58}$	Blue	Cluster 2
D (9,9)	$\sqrt{[(9-2.5)^2 + (9-3)^2]} = \sqrt{(42.25+36)} = \sqrt{78.25} = \mathbf{8.85}$	$\sqrt{[(9-9.5)^2 + (9-7.5)^2]} = \sqrt{(0.25+2.25)} = \sqrt{2.5} = \mathbf{1.58}$	Blue	Cluster 2
E (10,7)	$\sqrt{[(10-2.5)^2 + (7-3)^2]} = \sqrt{(56.25+16)} = \sqrt{72.25} = \mathbf{8.50}$	$\sqrt{[(10-9.5)^2 + (7-7.5)^2]} = \sqrt{(0.25+0.25)} = \sqrt{0.5} = \mathbf{0.71}$	Blue	Cluster 2
F (11,6)	$\sqrt{[(11-2.5)^2 + (6-3)^2]} = \sqrt{(72.25+9)} = \sqrt{81.25} = \mathbf{9.01}$	$\sqrt{[(11-9.5)^2 + (6-7.5)^2]} = \sqrt{(2.25+2.25)} = \sqrt{4.5} = \mathbf{2.12}$	Blue	Cluster 2

K-Means Clustering- Example

Assignment Summary Table:

Cluster	Points Assigned	Count
Cluster 1 (Red)	A, B	2
Cluster 2 (Blue)	C, D, E, F	4

Convergence Check: The assignments are identical to Iteration 1. The algorithm has converged.



How to Choose K – The Elbow Method

- **Inertia:** is the sum of squared distances from each data point to its assigned cluster center (centroid).

Formula:

$$\text{Inertia} = \sum_{\text{all points}} (\text{distance from point to its centroid})^2$$

- **Low inertia** → Points are very close to their cluster center → Good clustering
- **High inertia** → Points are far from their cluster center → Poor clustering



How to Choose K – The Elbow Method

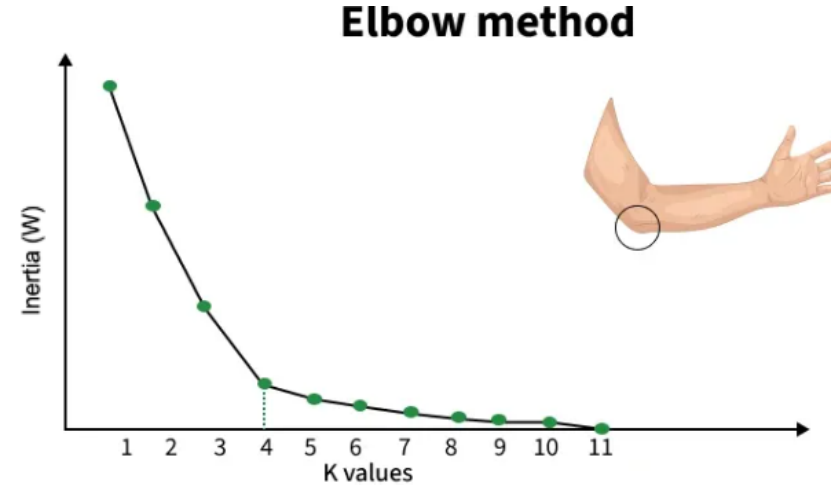
- Inertia always decreases as K increases. Adding more clusters allows points to be closer to their centroids, which reduces inertia. This is why inertia alone cannot determine the optimal number of clusters – it will always favor more clusters.

K	Inertia	Why
1	1000	All points forced into one cluster
2	600	Points can split into two tighter groups
3	420	Three groups are even tighter
4	400	Slight improvement
5	395	Very little improvement



How to Choose K – The Elbow Method

- Inertia is plotted against K to find the optimal number of clusters. The "elbow", where inertia stops decreasing sharply – indicates the best K.
- **Rule:** Choose the K where adding another cluster yields only a small reduction in inertia.



Model Interpretation Methods



Why Do We Need Interpretation Methods?

- **The Black Box Problem:** Many machine learning models (K-NN, Random Forest, Neural Networks) make accurate predictions but are impossible to understand by simply looking inside. You see inputs and outputs, but the reasoning remains hidden.
- **Why This is a Problem:**

Domain	Why Explanation is Required
Banking	Customer has right to know why loan was denied
Healthcare	Doctor cannot trust a diagnosis without reasoning
Legal	Regulations require explainable decisions (GDPR, etc.)
Business	Managers need to understand drivers to take action



Interpretation Methods

- Two Popular Interpretation Methods:

Method	Full Name	Core Idea
SHAP	SHapley Additive exPlanations	Uses game theory to fairly distribute credit for a prediction among all features
LIME	Local Interpretable Model-agnostic Explanations	Builds a simple, understandable model around each individual prediction



Evaluation Metrics for Classification



Classifier Evaluation

- Classifier evaluation refers to the process of assessing the performance of a classification model in machine learning. Various metrics and techniques are used to determine how well a classifier predicts the correct labels for given data. Below are the key classifier evaluation methods and metrics:

1. Confusion Matrix

2. Common Evaluation Metrics (Accuracy, Precision, Recall, F1-score)



Classifier Evaluation- Confusion Matrix

- A table that summarizes the performance of a classification model by comparing predicted vs. actual labels.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)



Classifier Evaluation- Common Evaluation Metrics

1. Accuracy

What it measures: Overall correctness of predictions.

Formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

When to use:

- Balanced datasets where FP and FN have similar costs.

Limitation: Misleading for imbalanced classes (e.g., 95% negative class).



Classifier Evaluation- Common Evaluation Metrics

2. Precision

What it measures: How many selected positives are truly correct.

Formula:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

When to use:

- Minimizing **false positives (FP)** is critical (e.g., spam detection, fraud prevention).



Not Spam for example and predicted as spam



Classifier Evaluation- Common Evaluation Metrics

3. Recall (Sensitivity, True Positive Rate)

What it measures: How many actual positives are correctly identified.

Formula:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

When to use:

- Minimizing **false negatives (FN)** is crucial (e.g., cancer detection, security threats).



The sample is cancer for example and predicted as normal



Classifier Evaluation- Common Evaluation Metrics

4. F1-Score

What it measures: Harmonic mean of precision and recall (balances both).

Formula:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

When to use:

- Imbalanced datasets where both FP and FN matter.
- Preferred over accuracy when class distribution is skewed.



Classifier Evaluation- Example

A dog vs. cat image classifier was tested on 100 images, producing this confusion matrix:

	Predicted Dog	Predicted Cat	Total
Actual Dog	70 (TP)	10 (FN)	80
Actual Cat	5 (FP)	15 (TN)	20
Total	75	25	100

Calculate:

- Accuracy
- Precision (for Dog class)
- Recall (for Dog class)
- F1-Score (for Dog class)



Classifier Evaluation- Example

- True Positives (TP) = 70
- False Negatives (FN) = 10
- False Positives (FP) = 5
- True Negatives (TN) = 15

a) Accuracy

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} = \frac{70 + 15}{100} = 0.85 \quad (85\%)$$

b) Precision (Dog class)

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{70}{70 + 5} = \frac{70}{75} \approx 0.933 \quad (93.3\%)$$



Classifier Evaluation- Example

c) Recall (Dog class)

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{70}{70 + 10} = \frac{70}{80} = 0.875 \quad (87.5\%)$$

d) F1-Score (Dog class)

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.933 \times 0.875}{0.933 + 0.875} \approx 0.903 \quad (90.3\%)$$



Evaluation Metrics for Regression



Fundamental Concept: Residual Error

- **Definition:** The residual (error) for observation i is the difference between the observed and predicted values.

$$\epsilon_i = y_i - \hat{y}_i$$

Where:

- y_i = actual (observed) value
- $\hat{y}_i$ = predicted value

Properties:

- Positive residual: Under-prediction ($y_i > \hat{y}_i$)
- Negative residual: Over-prediction ($y_i < \hat{y}_i$)
- Zero residual: Perfect prediction



Mean Absolute Error (MAE)

- MAE measures the average magnitude of prediction errors without considering direction. It is an L1-norm loss function. $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
- Given:** Three property valuation predictions.

Property	Actual Price (\$)	Predicted Price (\$)	Residual (\$)	Absolute Residual (\$)
A	300,000	310,000	-10,000	10,000
B	320,000	310,000	+10,000	10,000
C	310,000	315,000	-5,000	5,000

$$\text{MAE} = \frac{10,000 + 10,000 + 5,000}{3} = \frac{25,000}{3} = 8,333.33$$



Mean Squared Error (MSE)

- MSE computes the average squared difference between predicted and actual values. It is an L2-norm loss function.
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Property	Residual (\$)	Squared Residual (\$²)
A	-10,000	100,000,000
B	+10,000	100,000,000
C	-5,000	25,000,000

$$\text{MSE} = \frac{100,000,000 + 100,000,000 + 25,000,000}{3} = \frac{225,000,000}{3} = 75,000,000$$



Root Mean Squared Error (RMSE)

- RMSE is the square root of MSE, returning the error measure to the original units while preserving the quadratic penalty for large errors.

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- From the previous example:

$$\text{RMSE} = \sqrt{75,000,000} = 8,660.25$$



R² Score

- R² quantifies the proportion of variance in the dependent variable that is predictable from the independent variable(s). It represents the improvement of the model over a baseline that always predicts the mean.

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Where:

- $SS_{\text{res}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ (Residual Sum of Squares)
- $SS_{\text{tot}} = \sum_{i=1}^n (y_i - \bar{y})^2$ (Total Sum of Squares)
- $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ (Sample mean)

R² = (Error when you know nothing) - (Error when you know everything) divided by (Error when you know nothing)



R² Score

- Key Aspects of R² Performance Interpretation:

R ² Value	Interpretation	Action
0.90 – 1.00	Excellent fit	Deploy, but check for overfitting
0.70 – 0.89	Strong fit	Acceptable for most applications
0.50 – 0.69	Moderate fit	Consider additional features
0.30 – 0.49	Weak fit	Major improvements needed
0.00 – 0.29	Very weak	Reconsider model approach
< 0.00	Harmful	Reject model immediately

Negative R² occurs when your model produces worse predictions than simply using the mean as the prediction.



R^2 (Example 1: Simple Linear Regression (Good Model))

- A real estate analyst develops a model to predict house prices based on square footage. Evaluate the model using R^2 .

House	Actual Price (\$1000)	Predicted Price (\$1000)
1	150	155
2	180	178
3	220	215
4	250	248
5	300	305
6	350	342



R² (Example 1: Simple Linear Regression (Good Model))

Step 1: Calculate the Mean of Actual Values ($\bar{y}$)

$$\bar{y} = \frac{150+180+220+250+300+350}{6} = \frac{1,450}{6} = 241.67$$

Step 2: Calculate SS_{tot} (Total Sum of Squares)

$$SS_{tot} = \sum_{i=1}^n (y_i - \bar{y})^2$$

House	y_i	$\bar{y}$	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
1	150	241.67	-91.67	8,402.78
2	180	241.67	-61.67	3,802.78
3	220	241.67	-21.67	469.44
4	250	241.67	8.33	69.44
5	300	241.67	58.33	3,402.78
6	350	241.67	108.33	11,736.11

$$SS_{tot} = 8,402.78 + 3,802.78 + 469.44 + 69.44 + 3,402.78 + 11,736.11 = \mathbf{27,883.33}$$



R^2 (Example 1: Simple Linear Regression (Good Model))

Step 3: Calculate SS_{res} (Residual Sum of Squares)

$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

House	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	150	155	-5	25
2	180	178	2	4
3	220	215	5	25
4	250	248	2	4
5	300	305	-5	25
6	350	342	8	64

$$SS_{res} = 25 + 4 + 25 + 4 + 25 + 64 = \mathbf{147}$$



R^2 (Example 1: Simple Linear Regression (Good Model))

Step 4: Calculate R^2

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{147}{27,883.33}$$

$$R^2 = 1 - 0.00527 = 0.9947$$

$$R^2 = 0.995$$

The model explains 99.5% of the variance in house prices. This indicates an excellent fit, with only 0.5% of the total variation remaining unexplained.



R^2 (Example 2: Harmful Model (Negative R^2))

A flawed model predicts exam scores using an inversely correlated feature.

Student	Actual Score (%)	Predicted Score (%)
1	90	70
2	85	72
3	75	78
4	65	82
5	55	85
6	50	88



R² (Example 2: Harmful Model (Negative R²))

Step 1: Calculate $\bar{y}$

$$\bar{y} = \frac{90+85+75+65+55+50}{6} = \frac{420}{6} = 70$$

Step 2: Calculate SS_{tot}

Student	y_i	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
1	90	20	400
2	85	15	225
3	75	5	25
4	65	-5	25
5	55	-15	225
6	50	-20	400

$$SS_{tot} = 400 + 225 + 25 + 25 + 225 + 400 = \mathbf{1,300}$$



R^2 (Example 2: Harmful Model (Negative R^2))

Step 3: Calculate SS_{res}

Student	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	90	70	20	400
2	85	72	13	169
3	75	78	-3	9
4	65	82	-17	289
5	55	85	-30	900
6	50	88	-38	1,444

$$SS_{res} = 400 + 169 + 9 + 289 + 900 + 1,444 = \mathbf{3,211}$$



R² (Example 2: Harmful Model (Negative R²))

Step 4: Calculate R²

$$R^2 = 1 - \frac{3,211}{1,300} = 1 - 2.47 = -1.47$$

- The model exhibits negative R², indicating it performs substantially worse than simply predicting the mean score (70%).
- The negative coefficient suggests the model has learned an inverse relationship, producing predictions that deviate more from actual values than the baseline.
- **Recommendation:** Reject this model. Investigate feature engineering and direction of correlations.



Model Enhancement Methods



Ensemble Methods – Overview

- Ensemble methods combine multiple machine learning models (called "base learners") to produce a single, more accurate prediction than any individual model.
- **Types of Ensemble Methods**

Method	How It Works	Example Algorithms
Bagging	Train same model on random subsets of data, average predictions	Random Forest, Bagged Decision Trees
Boosting	Train models sequentially, each corrects previous errors	AdaBoost, XGBoost, Gradient Boosting
Voting	Combine different algorithm types, take majority or average	Hard Voting, Soft Voting
Stacking	Train a meta-model to learn best combination of base models	Stacked Generalization

Hyperparameter Optimization Using PSO or Genetic Algorithms



Hyperparameter Optimization

- Beyond feature selection, Particle Swarm Optimization and Genetic Algorithms are powerful tools for automatically finding optimal hyperparameters (K, distance metrics, weights, tolerance) for any machine learning model, often outperforming manual tuning and grid search.
- Hyperparameters are settings that control the learning process of a machine learning model. Unlike model parameters (which are learned from data), hyperparameters must be set by the user before training begins.
- The same PSO and GA methods you learned for feature selection can be directly applied to find optimal hyperparameters for any machine learning model.



Any Question?

